

Invariant Estimator Framework for Autonomous Cars

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I. STATE ON $SE_2(3)$

We define the state on the group manifold $SE_2(3)$, such that we can maintain invariant observer properties while keeping the system as observable as possible:

$$X \triangleq \begin{bmatrix} {}^{\mathcal{W}}R_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{p}_{\mathcal{B}} \\ \mathbf{0}_{3 \times 3} & I & 0 \\ \mathbf{0}_{3 \times 3} & 0 & I \end{bmatrix} \in SE_2(3) \quad (1)$$

$$\boldsymbol{\theta} = (\mathbf{b}_g, \mathbf{b}_a) \in \mathbb{R}^6$$

With the associated tangent vector defined as the following:

$$\boldsymbol{\xi} = [\boldsymbol{\xi}_{\phi} \quad \boldsymbol{\xi}_{\mathbf{v}} \quad \boldsymbol{\xi}_{\mathbf{p}} \quad \boldsymbol{\zeta}_g \quad \boldsymbol{\zeta}_a] \in \mathbb{R}^{15} \quad (2)$$

where the exponential on the state component of the tangent vector defined as $\exp([\boldsymbol{\xi}]^{\wedge}) = X^{-1}\hat{X}$, and $\boldsymbol{\zeta} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$.

II. PROCESS MODEL

We define the following continuous time (CT) dynamics:

$$\begin{aligned} {}^{\mathcal{W}}\dot{R}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\boldsymbol{\omega}^{\wedge} \\ {}^{\mathcal{W}}\dot{\mathbf{v}}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\mathbf{a} + {}^{\mathcal{W}}\mathbf{g} \\ {}^{\mathcal{W}}\dot{\mathbf{p}}_{\mathcal{B}} &= {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} \end{aligned} \quad (3)$$

And we know that these dynamics are *group-affine* - that is, that they yield a log-linear error which is bounded - this is further shown in the appendix. If we assume that the input measurements ${}^{\mathcal{B}}\boldsymbol{\omega}$ and ${}^{\mathcal{W}}\mathbf{a}_{\mathcal{B}}$ are constant over some small timestep interval $[t, t + \Delta t]$ (i.e., zero-order hold), then we know that the rotation matrix propagation is defined exactly:

$${}^{\mathcal{W}}R_{\mathcal{B}}(t_k + s) = {}^{\mathcal{W}}R_{\mathcal{B},k} \exp(s[\boldsymbol{\omega}]^{\wedge}) \quad (4)$$

This means that we can substitute this into the discretization of both velocity and position, with $s = u\Delta t$, for some timestep Δt . For instance, with velocity, the integrand yields:

$$\int_0^{\Delta t} \exp(s[\boldsymbol{\omega}]^{\wedge}) ds = \Delta t \int_0^1 \exp(u[\boldsymbol{\omega}]^{\wedge}) du = \Delta t \Gamma_1(\boldsymbol{\omega}) \quad (5)$$

And the integral behaves similarly for position. This means that after discretizing, we are left with the following dynamics that obey the $SO(3)$ manifold:

$$\begin{aligned} R_{k+1|k} &= R_{k|k} \Gamma_0(\boldsymbol{\omega}) \\ \mathbf{v}_{k+1|k} &= \mathbf{v}_{k|k} + {}^{\mathcal{W}}\mathbf{g}\Delta t + R_{k|k} \Gamma_1(\boldsymbol{\omega}) \mathbf{a} \Delta t \\ \mathbf{p}_{k+1|k} &= \mathbf{p}_{k|k} + \mathbf{v}_{k|k} \Delta t + \frac{1}{2} {}^{\mathcal{W}}\mathbf{g} \Delta t^2 + R_{k|k} \Gamma_2(\boldsymbol{\omega}) \mathbf{a} \Delta t^2 \\ \hat{\boldsymbol{\theta}}_{k+1|k} &= \hat{\boldsymbol{\theta}}_{k|k} \end{aligned} \quad (6)$$